Activity 3 — Retail Billing with Multiple Discount Rules

1. Problem Statement

Develop a Python program to calculate a customer's shopping bill by using product price, quantity, discount rules and tax. Display the subtotal, discount, tax and final bill amount.

2. Objectives

- To calculate the total price of products.
- To apply discount rules.
- To calculate tax and the final bill.
- To understand Python variables, arithmetic operators and conditional statements.

3. Assumptions

For this beginner-friendly example, assume:

- The customer enters the product price and quantity.
- If the subtotal is ₹1,000 or more, a 10% discount is applied.
- The maximum discount is ₹200.
- Tax is 5% of the amount after discount.
- Product price and quantity must be positive.

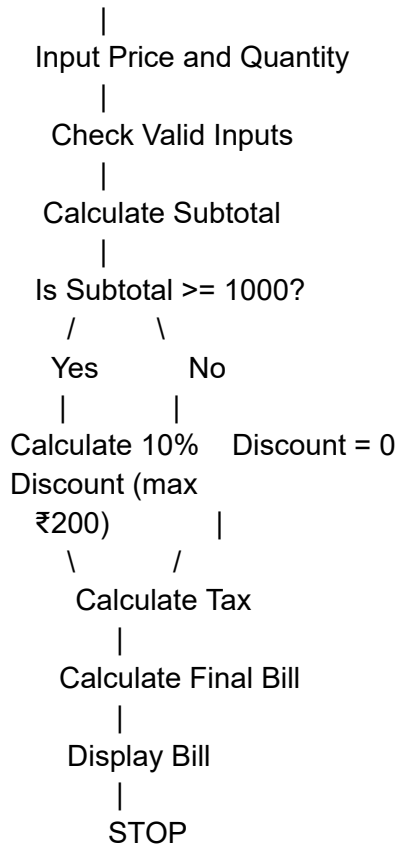
These are sample rules for learning purposes.

4. Algorithm

1. Start.
2. Read the product price and quantity.
3. Check whether the inputs are valid.
4. Calculate the subtotal.
5. If the subtotal is ₹1,000 or more, calculate a 10% discount, limited to ₹200.
6. Calculate tax at 5% on the amount after discount.
7. Calculate the final bill.
8. Display the subtotal, discount, tax and final bill.
9. Stop.

5. Flowchart

START



6. Pseudocode

```
START
INPUT price
INPUT quantity

IF price <= 0 OR quantity <= 0
   PRINT "Invalid input"
ELSE
   subtotal = price * quantity

   IF subtotal >= 1000
      discount = subtotal * 0.10
      IF discount > 200
         discount = 200
      ENDIF
   ELSE
      discount = 0
   ENDIF

   amount = subtotal - discount
   tax = amount * 0.05
   total = amount + tax

   PRINT subtotal
```

```
    PRINT discount
    PRINT tax
    PRINT total
ENDIF

STOP
```

7. Python Source Code

File name: `activity7.py`

```
price = float(input("Enter product price: "))
quantity = int(input("Enter quantity: "))

if price <= 0 or quantity <= 0:
    print("Invalid input")

else:
    subtotal = price * quantity

    if subtotal >= 1000:
        discount = subtotal * 0.10

        if discount > 200:
            discount = 200
    else:
        discount = 0

    amount = subtotal - discount
    tax = amount * 0.05
    total = amount + tax

    print("Subtotal: Rs.", subtotal)
    print("Discount: Rs.", discount)
    print("Tax: Rs.", tax)
    print("Final Bill: Rs.", total)
```

8. Test Cases

Product Price	Quantity	Expected Final Bill
₹500	3	₹1,417.50
₹200	2	₹420.00
₹1,000	3	₹2,940.00

-₹100	2	Invalid input

9. Output Screenshot and Conclusion

Sample Output:

Enter product price: 500
Enter quantity: 3
Subtotal: Rs. 1500.0
Discount: Rs. 150.0
Tax: Rs. 67.5
Final Bill: Rs. 1417.5

Output Screenshot: Run `activity7.py` in Python IDLE or VS Code. Capture the inputs and output, and save the screenshot as `activity7_output.png`.

Conclusion: This program calculates a retail bill by applying a discount and tax. It helps beginners understand arithmetic operations and conditional statements in Python.

Input code:

```
price = float(input("Enter product price: "))
quantity = int(input("Enter quantity: "))

if price <= 0 or quantity <= 0:
    print("Invalid input")

else:
    subtotal = price * quantity

    if subtotal >= 1000:
        discount = subtotal * 0.10

        if discount > 200:
            discount = 200
    else:
        discount = 0

    amount = subtotal - discount
    tax = amount * 0.05
    total = amount + tax

    print("Subtotal: Rs.", subtotal)
    print("Discount: Rs.", discount)
    print("Tax: Rs.", tax)
    print("Final Bill: Rs.", total)
```

Output:

```
Enter product price: 500
Enter quantity: 3
Subtotal: Rs. 1500.0
Discount: Rs. 150.0
Tax: Rs. 67.5
Final Bill: Rs. 1417.5

================================ RESTA
Enter product price: 200
Enter quantity: 2
Subtotal: Rs. 400.0
Discount: Rs. 0
Tax: Rs. 20.0
Final Bill: Rs. 420.0

================================ RESTA
Enter product price: 1000
Enter quantity: 3
Subtotal: Rs. 3000.0
Discount: Rs. 200
Tax: Rs. 140.0
Final Bill: Rs. 2940.0

================================ RESTA
Enter product price: -100
Enter quantity: 2
Invalid input
```